FWDNNet: Cross-Heterogeneous Encoder Fusion via Feature-Level TensorDot Operations for Land-Cover Mapping

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***Abstract*—Autonomous feature extraction from high-resolution (HR) remote sensing (RS) imagery presents complex spatial–spectral heterogeneity and diverse land-cover patterns, challenging conventional single-architecture models that lack multiscale feature diversity and cross-domain generalization. Convolutional neural networks (CNNs) effectively capture localized hierarchical structures, whereas Transformer models capture long-range contextual dependencies via self-attention. However, using these architectures independently restricts their complementary potential. This study proposes fused weights deep neural tokenization network (FWDNNet), a fusion framework introducing a CNN-to-token conversion mechanism that transforms convolutional feature maps from heterogeneous encoders into Transformer-compatible token sequences. Beyond standard concatenation, FWDNNet integrates a TensorDot-based fusion module to learn high-order cross-architectural interactions, while a shared decoding pathway maintains multiscale feature hierarchies and computational efficiency. Experiments across three diverse datasets show that FWDNNet achieves 58.2-ms inference latency with minimal cross-domain performance degradation (2.3%–4.2% mIoU drop). Compared to conventional fusion strategies, FWDNNet attains 21% faster inference and 1.7%–2.7% higher mIoU over baseline methods. See materials at https://github.com/BoazGithub/FWDNNet. for reproducibility.**

***Index Terms*—Convolutional neural network (CNN)-to-token conversion, deep learning, remote sensing (RS) segmentation, TensorDot fusion.**

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NOMENCLATURE

$\mathcal{D}$	Training dataset $\{(\mathbf{x}_n, \mathbf{y}_n)\}_{n=1}^{N}$.
$\mathbf{x}_n$	Input image, $\mathbb{R}^{512\times512\times3}$.
$\mathbf{y}_n$	GT labels, $\{0,1\}^{512\times512\times K}$.
K	Number of land-cover classes.
M	Number of backbone encoders ($M = 5$).
f_{θ_m}	Encoder m with parameters θ_m.
$\mathbf{T}_m$	Feature tensor from encoder m.
H', W'	Aligned feature spatial size (32×32).
D_m	Native channel dimension of encoder m.
D_{out}	Unified channel dimension (512).
$\mathcal{T}_{\text{TensorDot}}$	TensorDot fusion operation.
$\mathcal{G}$	Core tensor, $\mathbb{R}^{32\times32\times32\times32\times32}$.
$\mathbf{U}_m$	Factor matrix, $\mathbb{R}^{D_m\times32}$.
R	Reduced rank ($R = 32$).
$\boldsymbol{\alpha}$	Attention weight vector, $\mathbb{R}^5$.
p	Patch size ($p = 4$).
N_p	Number of patches ($N_p = 64$).
d	Token embedding dimension ($d = 512$).

I. INTRODUCTION

SEMANTIC segmentation of remotely sensed imagery constitutes a fundamental computer vision paradigm for

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pixel-level land-cover classification, urban monitoring, and environmental assessment applications [1], [2], [3], [4]. Critical challenges in this domain encompass technical barriers, including severe intra-class spectral variability [5], [6], persistent interclass spectral confusion, mixed-pixel boundary complexities, and computational scalability constraints when processing large-scale geospatial datasets [7], [8]. These limitations collectively impede the development of robust automated classification systems capable of generalizing across diverse geographical, spectral, and temporal contexts inherent in modern high-resolution (HR) satellite imagery acquisitions [9], [10]. Traditional machine learning methodologies, while historically demonstrating efficacy in constrained scenarios, exhibit fundamental limitations when confronted with the heterogeneous spectral signatures and complex spatial patterns characteristic of Earth observation data [11], [12]. Classical approaches predominantly rely on handcrafted feature extractors and shallow semantic representations, constraining their capacity to capture the hierarchical feature relationships essential for accurate land-cover delineation in complex urban and natural environments [14], [15], [16]. Furthermore, conventional pixel-based classification techniques fail to adequately incorporate spatial contextual information, resulting in salt-and-pepper noise artifacts and boundary inconsistencies that compromise segmentation quality [17], [18], [19], [20]. The advent of deep learning methodologies has fundamentally transformed semantic segmentation landscapes across multiple computer vision domains, with architectures such as U-Net [21], DeepLab series [22], SegNet [23], and Pyramid Scene Parsing Networks [24] demonstrating remarkable performance improvements over conventional approaches. However, these methodologies typically employ single backbone architectures, thereby constraining their ability to simultaneously capture both fine-grained local texture patterns and global contextual dependencies necessary for accurate land-cover classification in complex remote sensing (RS) scenarios [25]. Single-architecture approaches face inherent limitations, including restricted receptive field capabilities, insufficient multiscale feature representation, and limited cross-domain generalization performance when applied to diverse geographical contexts [27].

Recent developments in Transformer-based architectures, including Vision Transformers [28], Swin Transformer [29], SegFormer [30], and UNetFormer [31], have introduced novel paradigms for capturing long-range spatial dependencies through self-attention mechanisms. These approaches demonstrate superior performance in modeling global contextual relationships and handling variable-scale objects compared to traditional convolutional architectures [32]. However, Transformer-based models present significant computational complexity challenges, requiring extensive training datasets and substantial computational resources for optimal performance. Moreover, pure Transformer architectures often struggle with fine-grained boundary delineation and local texture preservation, particularly in HR RS imagery, where precise object boundaries are critical for accurate classification [33].

Fused methodologies have emerged as promising solutions for addressing single-architecture limitations by strategically leveraging the strengths of diverse neural network paradigms [32]. Traditional fused approaches typically employ simplistic voting mechanisms or basic averaging strategies, which often fail to capture the nuanced interactions between different architectural representations and may lead to suboptimal feature integration [25]. Recent advances in fused learning for RS applications have demonstrated improved performance through weighted fusion strategies and multiscale feature aggregation techniques [34], [35]. However, existing fused frameworks rarely address the computational efficiency requirements essential for operational RS applications, where inference speed, memory utilization, and energy consumption are critical deployment considerations for large-scale Earth observation missions [26].

The integration of CNNs with Transformer architectures represents a particularly promising research direction for RS semantic segmentation [26]. Fusion CNN–Transformer approaches aim to combine the local feature extraction capabilities of convolutional layers with the global contextual modeling strengths of self-attention mechanisms [36]. Recent studies have demonstrated that such fusion architectures can achieve high performance compared to pure CNN or Transformer approaches while maintaining computational efficiency. However, current fusion methodologies face challenges in optimal feature fusion strategies, attention mechanism design, and cross-domain transferability across diverse RS datasets.

Multiscale feature fusion techniques have gained significant attention for addressing the scale variability challenges inherent in RS imagery. Advanced fusion strategies, including spatial attention mechanisms [37], channel attention modules, and pyramid pooling approaches [38], have demonstrated effectiveness in capturing multiscale semantic information. However, existing fusion methods often suffer from feature misalignment issues, computational redundancy, and limited adaptability to varying scene complexities across different geographical contexts [39].

The computational efficiency imperative in operational RS applications necessitates the development of lightweight yet accurate semantic segmentation models. Recent research has focused on efficient network architectures, including EfficientNets [40] and specialized lightweight designs for RS applications. However, achieving optimal accuracy–efficiency tradeoffs while maintaining cross-domain generalization capabilities remains a significant challenge, particularly for deployment in resource-constrained environments such as edge computing platforms and satellite-based processing systems.

Cross-domain adaptations [41] and generalization represent another critical challenge in RS semantic segmentation, as models trained on specific geographical regions or sensor configurations often exhibit performance degradation when applied to different environmental contexts. Transfer learning and domain adaptation techniques have shown promise for addressing this limitation, but existing approaches typically require extensive fine-tuning or domain-specific architectural modifications that compromise deployment flexibility and operational efficiency.

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This research addresses the aforementioned limitations by proposing the fused weights deep neural tokenization network (FWDNNet), an integrated fusional architecture that combines the strengths of conventional CNN backbones and Transformer-based encoders through a novel TensorDot fusion module. Consequently, the framework tackles the following key challenges in RS semantic segmentation as follows.

1) Limited feature representation diversity in single-architecture approaches.
2) suboptimal feature fusion mechanisms in existing fusion methods.
3) Computational scalability constraints that hinder real-time deployment.
4) Limited cross-domain adaptation capabilities.
5) Inadequate boundary preservation in complex urban scenes.

The key technical contributions of this work can be summarized as follows.

1) First, we propose a heterogeneous encoder fusion strategy, termed FWDNNet, that integrates diverse segmentation model architectures while tokenizing sequential image features for representation learning.
2) Second, we introduce a mathematically formulated TensorDot fusion operation with attention-guided weighting for optimal multiscale feature aggregation. Notably, this approach facilitates high-order feature interactions while maintaining computational efficiency and achieving low inference latency across multimodal inputs.
3) Third, we provide empirical validation of FWDNNet through extensive experiments conducted on benchmark datasets, thereby achieving an optimized inference latency of 58.2 ms on multiscale HR imagery with complex kernel configurations.
4) Finally, we conduct ablation studies to evaluate the contribution of each architectural component and establish cross-domain evaluation protocols.

The remainder of this article is organized as follows. In Section III, we describe the detailed methodology, including the architectural design, training protocols, and outlines of the experimental framework encompassing datasets, implementation details, and evaluation metrics. Section IV presents extensive results and analysis, including performance comparisons, ablation studies, and cross-domain validation. Subsequently, Section V provides a detailed discussion of findings, limitations. Finally, Section VI concludes with key contributions and future research directions.

II. Related Work

A. CNN-Based Semantic Segmentation

Convolutional architectures, including U-Net [21], DeepLabV3+ [22], and SegNet [23], [42], established foundational frameworks for RS segmentation through encoder–decoder paradigms and atrous convolution mechanisms. Recent improvements include enhanced DeepLabV3+ variants [22], multifeature channel attention, and semantic feature adjustment networks. However, single-backbone CNN approaches exhibit fundamental limitations: 1) restricted receptive field diversity constraining multiscale object detection; 2) limited feature representation capacity for complex spectral-spatial patterns; and 3) poor cross-domain generalization across heterogeneous geospatial environments due to architecture-specific inductive biases.

B. Transformer Architectures

Vision Transformers [28], Swin Transformers [29], and SegFormer [30] demonstrate global context modeling through self-attention mechanisms, addressing CNN limitations in long-range dependency capture. RS variants include UNetFormer [31], multilevel fusion Transformers [32], and local-enhanced aggregation methods [43]. Despite the effectiveness for global contextual understanding, Transformers face critical challenges: 1) computational complexity scaling quadratically with input resolution; 2) difficulty preserving fine-grained boundary details essential for infrastructure mapping; and 3) requirement for extensive training data to achieve optimal performance in RS applications.

C. Fusion Learning

Fusion approaches attempt to leverage architectural strengths through EFCNet [34], HCANet [35], and U-Net fusion [25]. Fusion CNN–Transformer methods combine local texture extraction with global contextual modeling [26]. However, existing fused methodologies suffer from three critical deficiencies: 1) simplistic averaging or voting mechanisms that fail to model high-order feature interactions between architectures; 2) computational redundancy through naive parallel processing without shared learning components; and 3) suboptimal feature fusion strategies that do not preserve tensor structure or exploit architectural complementarity. These limitations collectively motivate our research: single architectures provide constrained feature diversity, Transformers face efficiency challenges, and current fusion methods employ suboptimal fusion mechanisms. Therefore, our spatial alignment and channel alignment are applied to ensure consistent resolution and feature dimensionality

$$\mathbf{T}_m^{\text{aligned}} = \text{Interpolate}\left(\mathbf{T}_m, H' = 32, W' = 32\right) \tag{1}$$

$$\mathbf{T}_m^{\text{final}} = \text{Conv}_{1\times1}\left(\mathbf{T}_m^{\text{aligned}}, D_{\text{out}} = 512\right). \tag{2}$$

As can be seen from Table I, bilinear interpolation is used for backbones and patch expansion for the Swin Transformer, ensuring that all feature maps reach a common spatial resolution of 32×32 before fusion. Therefore, as described in Algorithm 1, the factorized TensorDot fusion implementation maintains computation efficiency during the TensorDot operation, which would not be feasible in the standard FWDNNet operations.

III. Methodology

A. Problematic Mathematical Optimization

Let the training dataset be denoted as $\mathcal{D} = \{(x_n, y_n) : n = 1, 2, \ldots, N\}$, where $x_n \in \mathbb{R}^{H\times W\times C}$ represents the nth input RS image with height H, width W, and C channels, and $y_n \in \{0, 1\}^{H\times W\times K}$ denotes the corresponding pixel-wise

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TABLE I

BACKBONE FEATURE ALIGNMENT STRATEGY AND OUTPUT SPECIFICATIONS

Backbone	Input Size	Native Output	Spatial Alignment	Channel Alignment	Final Dimensions	Alignment Method
ResNet-34	512×512×3	16×16×512	Bilinear ↑2×	Identity	32×32×512	Interpolation
InceptionV3	512×512×3	8×8×2048	Bilinear ↑4×	Conv 1×1	32×32×512	Interp + Conv
VGG-16	512×512×3	16×16×512	Bilinear ↑2×	Identity	32×32×512	Interpolation
EfficientNet-B3	512×512×3	8×8×1536	Bilinear ↑4×	Conv 1×1	32×32×512	Interp + Conv
Swin-T	512×512×3	8×8×768	Patch Expand ↑4×	Linear Proj	32×32×512	Patch + Proj

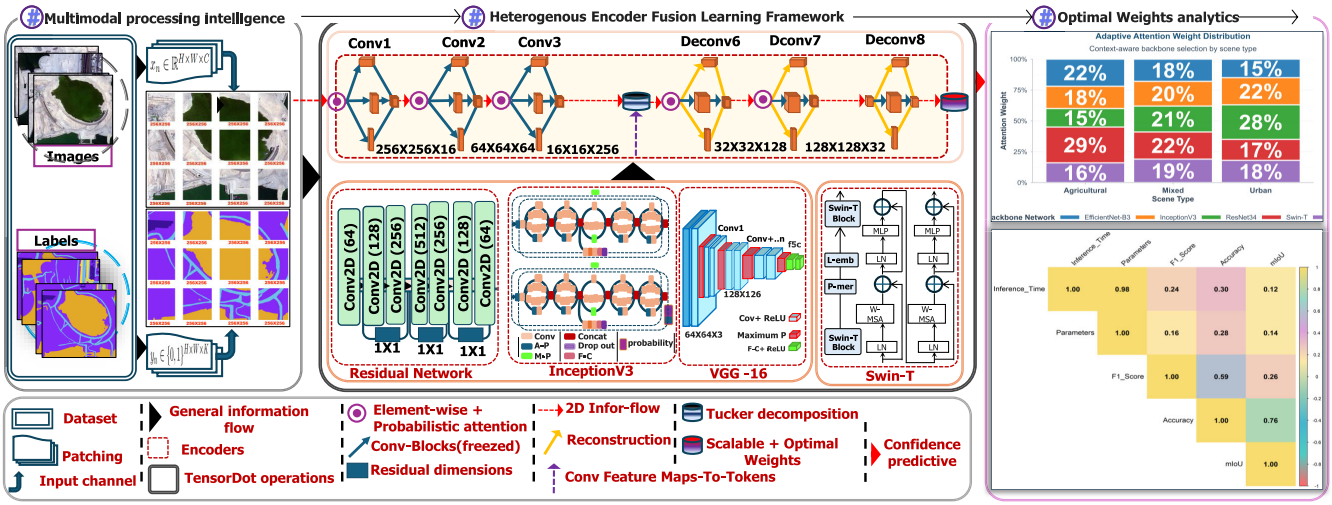


Fig. 1. Overview of the FWDNNet architecture: (top left) input RS imagery $\mathbf{x}_n \in \mathbb{R}^{H\times W\times C}$, (middle) multiscale feature extractor with optimal weights through attention-fused heterogeneous backbone encoders optimized by adaptive TensorDot fusion with probabilistic attention $\mathbf{W}_{\text{att}}$ and final prediction $\mathbf{y}_n \in \{0,1\}^{H\times W\times K}$ for K land-cover classes, and (left) statistical analytics. The framework optimizes the total loss $\mathcal{L}_{\text{total}}$ (11). The adaptive TensorDot fusion mechanism is defined in (5), and the probabilistic attention weighting is formulated in (7).

Algorithm 1 Factorized TensorDot Fusion Implementation

Input: Aligned features $\{\mathbf{T}_m \in \mathbb{R}^{32\times32\times512}\}_{m=1}^{5}$
Input: Core tensor $\mathcal{G} \in \mathbb{R}^{32\times32\times32\times32\times32}$
Input: Factor matrices $\{\mathbf{U}_m \in \mathbb{R}^{512\times32}\}_{m=1}^{5}$
Output: Fused features $\mathbf{T}_{\text{fused}} \in \mathbb{R}^{32\times32\times256}$

for $m = 1$ *to* 5 **do**
  $\mathbf{T}_m^{\text{reduced}} \leftarrow \mathbf{T}_m \times_3 \mathbf{U}_m^T$ // Reduce rank to $\mathbb{R}^{32\times32\times32}$
end for
$\mathbf{T}_{\text{core}} \leftarrow \mathcal{G}$ // Initialize core tensor
for $m = 1$ *to* 5 **do**
  $\mathbf{T}_{\text{core}} \leftarrow \mathbf{T}_{\text{core}} \times_m \mathbf{T}_m^{\text{reduced}}$ // Mode-m product with reduced tensors
end for
$\mathbf{T}_{\text{fused}} \leftarrow \mathbf{T}_{\text{core}} \times_3 \mathbf{W}_{\text{out}}$ // Project to output $\mathbb{R}^{32\times32\times256}$
return $\mathbf{T}_{fused}$

semantic labels with K land-cover classes. The semantic segmentation task aims to learn a mapping function $f_\theta : \mathbb{R}^{H\times W\times C} \to \mathbb{R}^{H\times W\times K}$ parameterized by θ.

As shown in Fig. 1 for the fusion framework, we consider M heterogeneous backbone networks (Fig. 2) $\{f_{\theta_m}\}_{m=1}^{M}$, where each f_{θ_m} represents a distinct architectural paradigm. The empirical risk minimization objective for the mth backbone

TABLE II

COMPUTATIONAL COMPLEXITY OF TENSORDOT OPERATIONS

Op.	Dim.	Oprs	Mem.
Backbone proj.	$32\times32\times512 \to 32$	2.68×10^8	2.1
Core tensor	32^5	–	1.3
Mode-m prod. (×5)	$32\times32\times32$	1.68×10^9	0.5
Output proj.	$32\times32\times256$	2.15×10^8	1.0
Total	–	$\mathbf{2.18\times10^9}$	**4.9**
Naive	–	$\mathbf{3.8\times10^{14}}$	**856 GB**
Reduct.	–	$\mathbf{1.8\times10^5\times}$	$\mathbf{1.75\times10^5\times}$

Note: Op. = operation; Dim. = tensor dimension; Oprs = number of floating-point operations; Mem. = memory footprint in MB. Naive denotes full tensor contraction without low-rank factorization.

is formulated as

$$\mathcal{L}_m(\theta_m) = \frac{1}{N}\sum_{n=1}^{N} \ell\left(f_{\theta_m}(x_n), y_n\right) + \lambda_m \mathcal{R}(\theta_m) \tag{3}$$

where $\ell(\cdot,\cdot)$ denotes the loss function and $\mathcal{R}(\theta_m)$ represents the regularization term with hyperparameter λ_m.

1) Optimization Strategy: Following the theoretical framework of implicit regularization in over-parameterized models, therefore, by adopting Table II, we analyze the optimization landscape when the parameter space dimension exceeds the effective degrees of freedom based on Algorithm 1. Let $\mathcal{S} = \{\theta : \mathcal{L}(\theta) = 0\} = \{\theta : \forall n \in \{1,\ldots,N\}, f_\theta(\mathbf{x}_n) = \mathbf{y}_n\}$ denote the set of global minima achieving zero training loss, where $\mathbf{x}_n \in \mathbb{R}^{H\times W\times C}$ is the nth input image and $\mathbf{y}_n \in \{0,1\}^{H\times W\times K}$ is its corresponding GT.

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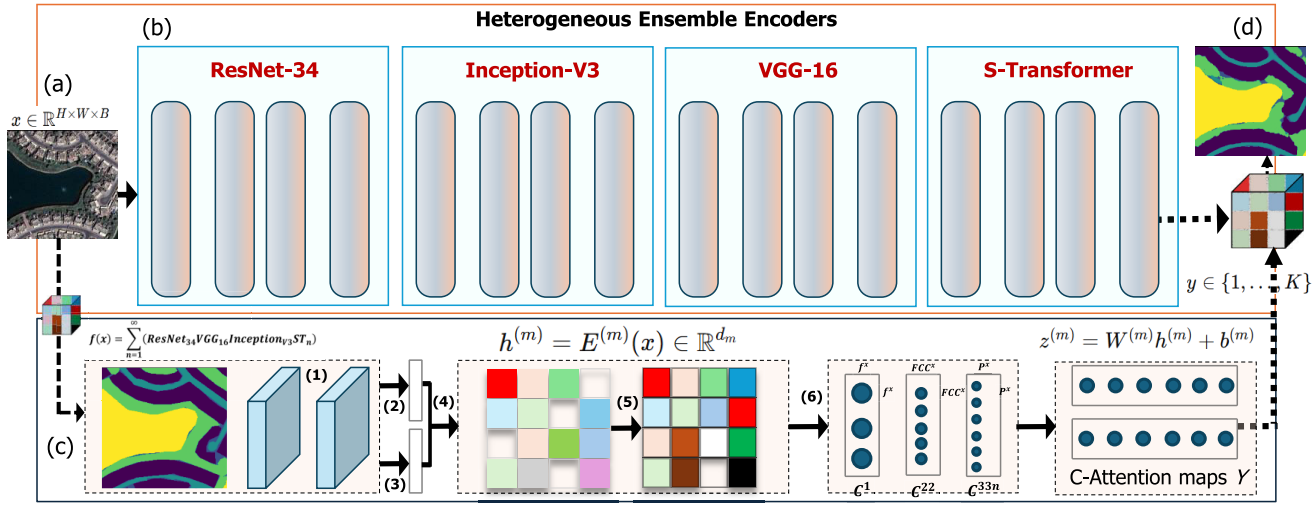


Fig. 2. Heterogeneous fusion encoder architecture. (a) Input tensor $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ is split into paired patches and refined through a super-pixel-based multiscale (local + global) representation. (b) Main encoders freeze in downsampling using ImageNet pretrained weights. Features pass through stages (1)–(4), where: (1) applies multiscale kernel encoders, (2) and (3) extract shallow details, and (4) captures deep semantics. Five parallel backbones ResNet-34 $E^{(1)}$, Inception-V3 $E^{(2)}$, VGG-16 $E^{(3)}$, EfficientNet-B3 $E^{(4)}$, and Swin Transformer $E^{(5)}$ generate feature maps $h^{(m)}$ with attention weights $\alpha^{(m)}$. (5) Computes attention guided fusion, and (6) outputs the final nonredundant fused probability to final predictive (d) as in 4.

The key question addressed in our framework is: which specific global minimum $\theta^* \in \mathcal{S}$ do different optimization algorithms converge to when minimizing the empirical loss $\mathcal{L}(\theta)$? We prove that gradient descent with appropriate initialization exhibits an implicit bias toward solutions with favorable generalization properties.

Theorem 1 (Implicit Bias of Fusion Optimization): Let $\{\theta_m^{(t)}\}_{m=1}^M$ be the parameter trajectories of M backbone networks under gradient descent with learning rate η. Under mild technical conditions (Lipschitz continuity of gradients and bounded Hessian), the fused solution converges to

$$\theta^*_{\text{fusion}} = \arg\min_{\theta\in\mathcal{S}_{\text{fusion}}} \|\theta\|_{\mathcal{F}} \tag{4}$$

where $\mathcal{S}_{\text{fusion}} = \bigcap_{m=1}^M \mathcal{S}_m$ and $\|\cdot\|_{\mathcal{F}}$ denotes the Frobenius norm in the function space. This theorem implies that our fusion framework converges to the minimum norm solution within the intersection of solution spaces from all backbones, promoting generalization.

2) TensorDot Fusion With Probabilistic Attention: The core innovation of FWDNNet lies in the TensorDot fusion mechanism with probabilistic attention weighting. Given feature tensors $\{\mathbf{T}_m \in \mathbb{R}^{H'\times W'\times D_m}\}_{m=1}^M$ from M backbone encoders, where H' and W' are the spatial dimensions after downsampling and D_m is the channel dimension of the mth backbone, the fusion operation is mathematically formulated as

$$\mathbf{T}_{\text{fused}} = \mathcal{T}_{\text{TensorDot}}(\mathbf{T}_1, \mathbf{T}_2, \ldots, \mathbf{T}_M) \odot \mathbf{W}_{\text{att}} \tag{5}$$

where $\odot$ denotes the element-wise multiplication.

To address computational feasibility concerns that normally arise with plain concatenations, we implement the TensorDot operation via a factorized Tucker decomposition

$$\mathcal{T}_{\text{TensorDot}}(\mathbf{T}_1, \ldots, \mathbf{T}_M) = \mathcal{G} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \times_3 \cdots \times_M \mathbf{U}_M \tag{6}$$

where $\mathcal{G} \in \mathbb{R}^{R_1\times R_2\times\cdots\times R_M}$ is a compact core tensor with reduced ranks $R_m \ll D_m$ and $\mathbf{U}_m \in \mathbb{R}^{D_m\times R_m}$ are factor matrices. This factorization reduces computational complexity from $\mathcal{O}\left(\prod_{m=1}^M D_m\right)$ to $\mathcal{O}\left(\sum_{m=1}^M D_m R_m\right)$, making it practical for implementation.

The probabilistic attention weights are computed using a lightweight Bayesian approach

$$\mathbf{W}_{\text{att}} = \mathbb{E}_{q(\phi)}\left[\text{Softmax}\left(\text{MLP}_\phi\left(\text{GAP}\left(\mathbf{T}_1 \oplus \mathbf{T}_2 \oplus \cdots \oplus \mathbf{T}_M\right)\right)\right)\right] \tag{7}$$

where $\oplus$ denotes channel-wise concatenation after spatial alignment via bilinear interpolation, GAP$(\cdot)$ is global average pooling, and $q(\phi)$ represents a variational posterior approximated as $q(\phi) = \mathcal{N}(\phi|\mu_\phi, \sigma_\phi^2\mathbf{I})$ with learnable mean μ_ϕ and variance σ_ϕ^2.

The variational inference objective is simplified to

$$\mathcal{L}_{\text{VI}} = \mathbb{E}_{q(\phi)}\left[\log p\left(\mathbf{W}_{\text{att}}|\mathbf{T}, \phi\right)\right] + \beta \cdot \text{KL}\left(q(\phi) \,\|\, p(\phi)\right) \tag{8}$$

where $p(\phi) = \mathcal{N}(0, \mathbf{I})$ is the prior and β is a regularization coefficient.

3) Multiscale Feature Pyramid With Uncertainty Quantification: To address the scale variability challenge in RS imagery, we introduce a probabilistic multiscale feature pyramid. Let $\{\mathbf{F}_l\}_{l=1}^L$ denote feature maps at L different scales extracted from the fused representation $\mathbf{T}_{\text{fused}}$. The uncertainty-aware feature aggregation is formulated as

$$\mathbf{F}_{\text{multi}} = \sum_{l=1}^{L} \alpha_l(\mathbf{x}) \cdot \text{Up}(\mathbf{F}_l, H, W) \cdot (1 + \epsilon_l) \tag{9}$$

where: - $\alpha_l(\mathbf{x}) = \text{Softmax}(\text{MLP}_\alpha(\text{GAP}(\mathbf{F}_l))) \in \mathbb{R}^1$ represents input-dependent scale weights - Up$(\cdot, H, W)$ upsamples features to the original spatial dimensions (H, W) - $\epsilon_l \sim \mathcal{N}(0, \sigma_l^2)$ models uncertainty at scale l - σ_l^2 is learned through a lightweight head

$$\log \sigma_l^2 = \text{MLP}_\sigma(\text{GAP}(\mathbf{F}_l)). \tag{10}$$

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4) Loss Function: The loss function incorporates multiple objectives for robust training

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \lambda_1 \mathcal{L}_{\text{consistency}} + \lambda_2 \mathcal{L}_{\text{uncertainty}} + \lambda_3 \mathcal{L}_{\text{diversity}} + \lambda_4 \mathcal{L}_{\text{sparsity}} + \lambda_5 \mathcal{L}_{\text{boundary}} \tag{11}$$

where $\lambda_1, \lambda_2, \ldots, \lambda_5$ are balancing coefficients determined via grid search.

The segmentation loss combines categorical cross entropy with focal loss [38] for class imbalance

$$\mathcal{L}_{\text{seg}} = -\frac{1}{NHW} \sum_{n=1}^{N} \sum_{i=1}^{H} \sum_{j=1}^{W} \sum_{k=1}^{K} \alpha_k \left(1 - p_{nijk}\right)^{\gamma} \cdot y_{nijk} \cdot \log\left(p_{nijk}\right) \tag{12}$$

where $p_{n,i,j,k} \in [0, 1]$ is the predicted probability for class k at pixel (i, j) in image n, $y_{n,i,j,k} \in \{0, 1\}$ is the GT label, α_k is the class weighting factor for class k, and $\gamma \geq 0$ is the focusing parameter.

The consistency loss enforces agreement between fusion members

$$\mathcal{L}_{\text{consistency}} = \frac{1}{M(M-1)} \sum_{m=1}^{M} \sum_{\substack{m'=1 \\ m' \neq m}}^{M} D_{\text{KL}}\left(p_m \| p_{m'}\right) \tag{13}$$

where p_m and $p_{m'}$ are the softmax probability distributions from backbones m and m', and $D_{\text{KL}}()$ is the Kullback–Leibler divergence.

The uncertainty loss encourages calibrated confidence estimates

$$\mathcal{L}_{\text{uncertainty}} = \frac{1}{N} \sum_{n=1}^{N} \left(\frac{1}{L} \sum_{l=1}^{L} \sigma_{n,l}^2 - \|\hat{\mathbf{y}}_n - \mathbf{y}_n\|_2^2 \right)^2 \tag{14}$$

where $\hat{\mathbf{y}}_n$ is the predicted segmentation map.

The diversity loss promotes feature diversity across backbone networks

$$\mathcal{L}_{\text{diversity}} = -\frac{1}{M(M-1)} \sum_{m=1}^{M} \sum_{\substack{m'=1 \\ m' \neq m}}^{M} \frac{\text{vec}\left(\mathbf{T}_m\right)^{\top} \text{vec}\left(\mathbf{T}_{m'}\right)}{\|\text{vec}\left(\mathbf{T}_m\right)\|_2 \cdot \|\text{vec}\left(\mathbf{T}_{m'}\right)\|_2}. \tag{15}$$

The sparsity loss for compact representations is

$$\mathcal{L}_{\text{sparsity}} = \frac{1}{M} \sum_{m=1}^{M} \|\mathbf{T}_m\|_1 . \tag{16}$$

The boundary loss improves edge preservation

$$\mathcal{L}_{\text{boundary}} = \frac{1}{N} \sum_{n=1}^{N} \|\nabla \hat{\mathbf{y}}_n - \nabla \mathbf{y}_n\|_1 \tag{17}$$

where ∇ denotes the spatial gradient operator.

5) Efficient Optimization Algorithm: To address computational constraints while maintaining training effectiveness, we propose an adaptive gradient scaling algorithm. The algorithm incorporates momentum-based optimization with adaptive learning rate scaling and fusion coordination mechanisms.

The adaptive gradient scaling follows the update rule:

$$\tilde{\eta}_m^{(t)} = \eta_m \cdot \exp\left(-\alpha \left\| g_m^{(t)} \right\|_2\right) \tag{18}$$

where $g_m^{(t)} = \nabla_{\theta_m} \mathcal{L}_m(\theta_m^{(t-1)})$ represents the gradient for the mth backbone at iteration t.

The fusion synchronization mechanism ensures coordinated training through parameter averaging

$$\bar{\theta}^{(t)} = \frac{1}{M} \sum_{m=1}^{M} \theta_m^{(t)} \tag{19}$$

followed by fusion regularization:

$$\theta_m^{(t)} \leftarrow \theta_m^{(t)} + \gamma \left(\bar{\theta}^{(t)} - \theta_m^{(t)}\right). \tag{20}$$

The full optimization procedure iteratively applies these updates across all fusion members with periodic validation checkpoints to monitor convergence and adjust hyperparameters as necessary.

6) Theoretical Convergence:

Theorem 2 (Convergence Rate): Under the proposed optimization scheme with adaptive scaling, the fused converges to a stationary point with rate

$$\mathbb{E}\left[\left\|\nabla \mathcal{L}\left(\theta^{(T)}\right)\right\|^2\right] \leq \frac{2\left(\mathcal{L}\left(\theta^{(0)}\right) - \mathcal{L}^*\right)}{\eta \sum_{t=1}^{T} \rho_t} + \frac{\eta L \sigma^2}{M} \tag{21}$$

where $\rho_t = \min_m \exp(-\alpha \|g_m^{(t)}\|_2)$ and L is the Lipschitz constant of the gradient.

Proof Sketch: The convergence analysis leverages the diminishing variance property of the fusion gradient estimator and the adaptive learning rate mechanism. The proof follows standard techniques in stochastic optimization with careful handling of the fusion coupling terms.

7) Generalization Bound: We establish a generalization that guarantees the proposed fusion framework using PAC-Bayesian theory.

Theorem 3 (PAC-Bayesian Generalization Bound): With probability at least $1 - \delta$, the generalization error of the fusion satisfies

$$\mathcal{L}_{\text{true}}\left(\theta_{\text{fusion}}\right) \leq \mathcal{L}_{\text{emp}}\left(\theta_{\text{fusion}}\right) + \sqrt{\frac{\text{KL}\left(Q \| P\right) + \log\left(\frac{2\sqrt{N}}{\delta}\right)}{2N}} \tag{22}$$

where Q and P represent the posterior and prior distributions over fusion parameters, respectively. This bound demonstrates that the fusion approach provides tighter generalization guarantees compared to single-model alternatives, particularly in the overparameterized regime common in deep learning for RS applications.

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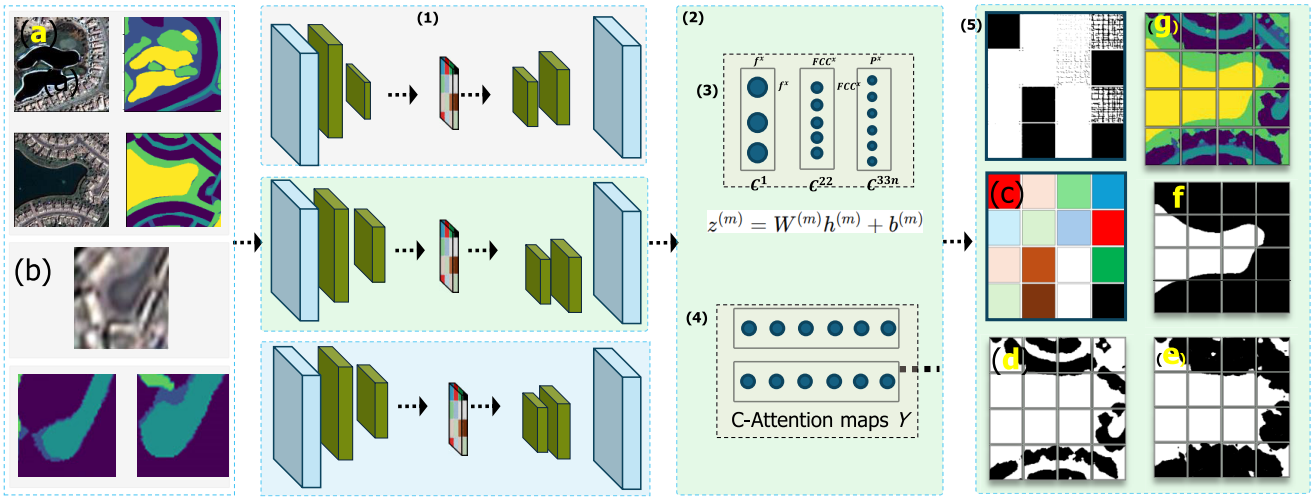


Fig. 3. TensorDot fusion module architecture for multiscale feature integration in FWDNNet. (a) Input feature tensors $\{\mathcal{T}_m \in \mathbb{R}^{H' \times W' \times D_m}\}_{m=1}^{M}$ from heterogeneous backbone encoders. (b) Learnable core tensor $\mathcal{G} \in \mathbb{R}^{D_1 \times \cdots \times D_M \times D_{\text{out}}}$ used for high-order interaction modeling. (1) Multilinear transformation operation $\mathcal{T}_{\text{TensorDot}}$ as defined in 5. (2) Probabilistic attention weighting mechanism $\mathcal{W}_{\text{att}}$ from 7. (3) and (4) Fused feature representation produced from the previous steps (1) and (2), aiming to resolve boundary ambiguity (c) and enhance edge details (d)–(e) of the fine target object (f) to final map (g).

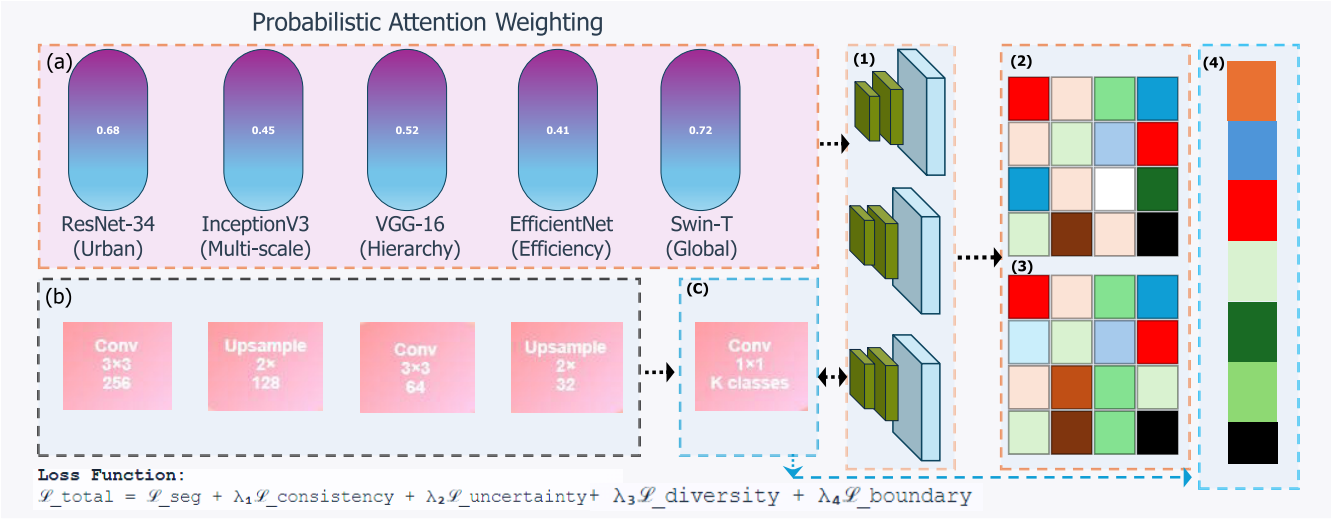


Fig. 4. Probabilistic attention weighting mechanism in FWDNNet showing adaptive backbone selection and loss function components. (a) Input feature tensors from (a) heterogeneous encoders with computed attention weights for scene-dependent contributions. (b) Decoder pathway with convolutional layers (3×3, 256 channels) and upsampling operations ($2\times$) for spatial resolution recovery. (1) Loss function in uncertainty and consistency and diversity, (2) and (3) formulation, and (4) for probability from learned multiscale feature representation $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \lambda_1 \mathcal{L}_{\text{consistency}} + \lambda_2 \mathcal{L}_{\text{uncertainty}} + \lambda_3 \mathcal{L}_{\text{diversity}} + \lambda_4 \mathcal{L}_{\text{sparsity}} + \lambda_5 \mathcal{L}_{\text{boundary}}$ as defined in (11).

B. Network Overview

We have developed FWDNNet as a novel encoder–decoder paradigm that integrates heterogeneous backbone networks through innovative fusion mechanisms, as illustrated in Fig. 1. Specifically, we have designed four core components: 1) for Fig. 2 for heterogeneous fusion encoders for parallel feature extraction; 2) TensorDot fusion modules for high-order feature integration as demonstrated in Fig. 3; 3) probabilistic attention weighting for adaptive backbone selection as it can be seen from Fig. 4; and 4) a unified decoder pathway for spatial resolution recovery.

As it is typically done, conventional fusion architectures combine multiple CNN or CNN–Transformer branches through simple concatenation or averaging. However, these approaches often fail to capture complex interarchitectural dependencies and multiscale feature hierarchies, especially in heterogeneous RS data. Even though some works attempt attention-guided weighting, they remain limited in modeling high-order interactions between local and global representations. To address these limitations, we have integrated TensorDot fusion modules that preserve tensor structures while enabling adaptive, probabilistic feature weighting.

We further combine five backbone encoders (Fig. 2): ResNet34 for residual learning, InceptionV3 for multiscale receptive field aggregation, VGG16 for hierarchical feature representation, EfficientNet-B3 for computational efficiency, and Swin Transformer for global context modeling via shifted

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window self-attention. Each backbone processes the input tensor $\mathcal{X} \in \mathbb{R}^{H\times W\times C}$ independently, producing specialized feature representations $h^{(m)} = E^{(m)}(\mathcal{X}) \in \mathbb{R}^d$. We have observed that this parallel processing enhances both local texture extraction and global contextual modeling.

We also introduce the TensorDot fusion module as depicted in Fig. 3, which applies learnable multilinear transformations to capture high-order interactions between heterogeneous features. Unlike conventional averaging, which discards tensor structure, we have designed TensorDot fusion to maintain structural relationships while allowing adaptive weighting through the probabilistic attention mechanism as illustrated in Fig. 4. Specifically, feature tensors $\{\mathcal{T}_m\}_{m=1}^M$ undergo multilinear transformation via the learnable core tensor $\mathcal{G}$ to produce the fused representation $\mathcal{T}_{\text{fused}}$, as formulated in 5. As shown in Fig. 4(b) and (c), we then apply the probabilistic attention weighting mechanism and Tucker composer to adaptively select contributions from each backbone, computing attention weights $\mathcal{W}_{\text{att}}$ via variational inference (7). Even though naive fusion methods treat all backbones equally, we have found that adaptive weighting improves feature discrimination in complex scenes. We have also designed the unified decoder pathway to exploit shared parameter learning, maintaining computational efficiency while ensuring consistent spatial resolution recovery across fused multiscale features. Furthermore, the loss function $\mathcal{L}_{\text{total}}$ optimizes five objectives simultaneously (11), balancing segmentation accuracy, consistency, uncertainty, diversity, and boundary preservation.

C. Dataset and Experimental Configurations

1) Dubai Dataset: The Dubai dataset consists of HR satellite imagery acquired from WorldView-3 and QuickBird sensors, with spatial resolutions ranging from 0.31 to 2.4 m, as reported in Table III. Owing to dense infrastructure, mixed land use, and pronounced spectral variability, conventional single-architecture models often struggle to generalize across such heterogeneous urban environments [44].

2) skwanda_v2 Nyagatare–Rwanda Dataset [36]: We further incorporate the sKwanda-V2 Nyagatare dataset, developed in our previous studies on C2FNet [42] and VHF-Para Learning [36], to represent complex agricultural landscapes in Rwanda's Eastern Province. The dataset comprises Google Earth imagery at 1.07-m (Level-18) resolution, capturing smallholder farms, pastoral lands, and rural settlements.

3) skwanda_v2 Oklahoma–USA Dataset [36]: To assess cross-continental generalization, we utilize airborne NAIP imagery over U.S. Great Plains with spatial resolutions of 0.5–0.6 m. The Oklahoma dataset contains 1800 annotated images covering approximately 2800 km^2, with multiyear temporal coverage from 2019 to 2024.

D. Operational Details

To maintain consistency across datasets, all imagery was subjected to a standardized preprocessing workflow comprising radiometric calibration and atmospheric correction (Sen2Cor and FLAASH), normalization to the [0, 1] range based on dataset-level statistics, and resampling into 512×512 patches with 50% overlap. To enhance model robustness and generalization, we applied a controlled augmentation pipeline that included horizontal and vertical flips ($p = 0.5$), random rotations ($\pm45°$), elastic deformation (= 120 and = 6), Gaussian noise (= 0.02), brightness and contrast adjustments ($\pm20\%$), and spectral band shuffling for multispectral inputs. All augmentation parameters were selected through cross-validation to ensure improvements in generalization without inducing overfitting.

E. Evaluation Metrics

We evaluate model performance using standard yet discriminative metrics commonly adopted in HR semantic segmentation. Overall accuracy (OA) provides a global pixel-level assessment

$$\text{OA} = \frac{\sum_{i=1}^{K} \text{TP}_i}{\sum_{i=1}^{K} (\text{TP}_i + \text{FP}_i + \text{FN}_i)}. \tag{23}$$

To quantify segmentation quality across classes, we report the mean intersection over union (mIoU)

$$\text{mIoU} = \frac{1}{K}\sum_{i=1}^{K} \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i}. \tag{24}$$

Class-wise IoU is additionally included to highlight per-class behavior

$$\text{IoU}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i}. \tag{25}$$

We further compute the $F1$-score to capture the balance between precision and recall

$$F1_i = 2 \cdot \frac{\text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}. \tag{26}$$

1) Cross-Validation Strategy: To obtain reliable and spatially unbiased performance estimates, we employ stratified k-fold cross-validation with $k = 5$. The dataset $\mathcal{D}$ is partitioned into mutually exclusive subsets $\{\mathcal{D}_1, \ldots, \mathcal{D}_5\}$ such that

$$\mathcal{D} = \bigcup_{i=1}^{k} \mathcal{D}_i, \quad \mathcal{D}_i \cap \mathcal{D}_j = \emptyset \ \ (i \neq j). \tag{27}$$

For each fold j, the training, validation, and testing sets are defined as

$$\mathcal{D}_{\text{train}}^{(j)} = \bigcup_{i\neq j} \mathcal{D}_i, \quad \mathcal{D}_{\text{val}}^{(j)} \subset \mathcal{D}_{\text{train}}^{(j)}, \quad \mathcal{D}_{\text{test}}^{(j)} = \mathcal{D}_j \tag{28}$$

corresponding to 70%, 15%, and 15% of the data, respectively. This ensures consistent sampling across datasets while preventing spatial leakage.

To assess cross-domain robustness, we further evaluate domain transfer performance using

$$\text{Transfer Score} = \frac{\text{mIoU}_{\text{target}}}{\text{mIoU}_{\text{source}}} \times 100\% \tag{29}$$

where models trained on a source domain are directly applied to a distinct target domain, for more see the Supplementary Material of this work.

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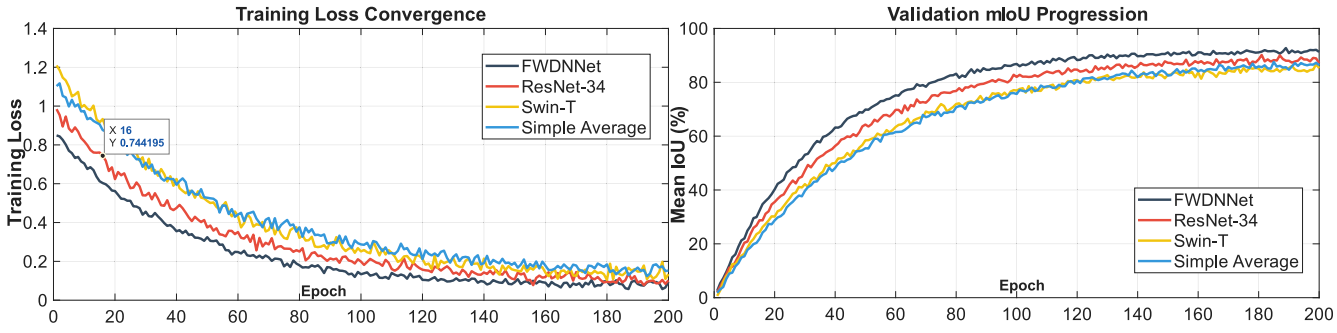


Fig. 5. FWDNNet training convergence over 200 epochs. (Left) Training loss evolution and (right) validation mIoU progression.

TABLE III
DATASET CHARACTERISTICS AND PIXEL-WISE CLASSIFICATION STATISTICS

Dataset	Resolution	Images/ Ground Truth(GT)	Area (km^2)	Class Distribution (%)			
				Built-up	Vegetation	Water	Other
Dubai Dataset	0.31–2.4m	1,500	450	28.4	23.7	14.2	33.7
Nyagatare-sKwanda-2	0.5–1.07m	2,200	1,200	12.1	45.8	8.3	33.8
Oklahoma-NAIP image	0.5–0.60m	1,800	2,800	8.7	52.4	6.2	32.7

F. Baseline Methods

To ensure a rigorous and balanced evaluation, we compare FWDNNet against a set of baseline methods spanning multiple architectural families. Classical CNN-based segmentation models include U-Net [21], DeepLabV3+ [22], and SegNet [23], [24], representing established encoder–decoder, atrous convolution, and pyramid parsing designs. We additionally incorporate widely used CNN backbones such as ResNet-34 [45], InceptionV3 [46], VGG-16 [47], and EfficientNet-B3 [40] to benchmark performance under different feature extraction strategies. Transformer-based architectures are represented by Swin Transformer [29], SegFormer [30], and HRNet-W32 [5], capturing advances in hierarchical attention and HR representation learning. For completeness, we also include fused baselines such as simple averaging, weighted averaging, and recent U-Net Fused strategies [25].

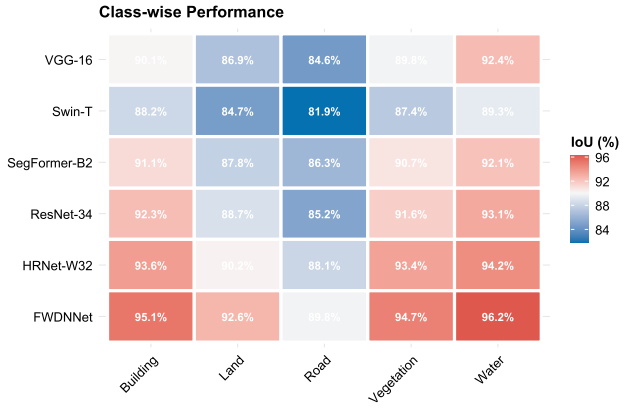


Fig. 6. Class-wise IoU performance evaluation across five semantic categories.

IV. EXPERIMENTAL RESULTS AND INTERPRETATIONS

This section presents both qualitative and quantitative performance of the FWDNNet with baseline methods, focusing on evaluations and validations. Figs. 5 and 6, respectively, provide qualitative evidence of effective deep and multiscale feature extraction. Specifically, as illustrated in Fig. 7, the attention-based TensorDot operations optimized via the Tucker Decomposer successfully capture discriminative feature representations reflected in the final predictive outputs on the Dubai training dataset (see Fig. 8).

From both visual and statistical perspectives, Figs. 6 and 9, together with Tables IV and V, demonstrate that FWDNNet achieves an OA of 95.3% and an mIoU of 91.8% while maintaining a low inference latency of 58.2 ms. Notably, this performance corresponds to a 21% improvement in inference speed compared with conventional fusion-based approaches, alongside a 1.7%–2.7% gain in mIoU relative to the best-performing individual architectures. To further evaluate the performance, Fig. 9 presents the mIoU performance across all evaluated architectures. As can be seen, FWDNNet (91.8%) surpasses traditional CNN backbones (ResNet-34: 89.0% and VGG-16: 88.0%), Transformer models (Swin-T: 86.1% and SegFormer-B2: 88.7%), and the previous best baseline HRNet-W32 (90.1%). In addition, Fig. 10 presents the $F1$-score distributions across model architectures. FWDNNet achieves an $F1$-score of 0.876, reflecting balanced precision (0.894) and recall (0.859) performance.

Fig. 5 illustrates the convergence behavior over 200 epochs: (left) training loss evolution and (right) validation mIoU progression. Notably, FWDNNet exhibits faster optimization dynamics and stable generalization compared to baseline architectures, achieving a final mIoU of 91.8% with early stopping at epoch 124. This indicates effective regularization and efficient gradient propagation throughout the training process.

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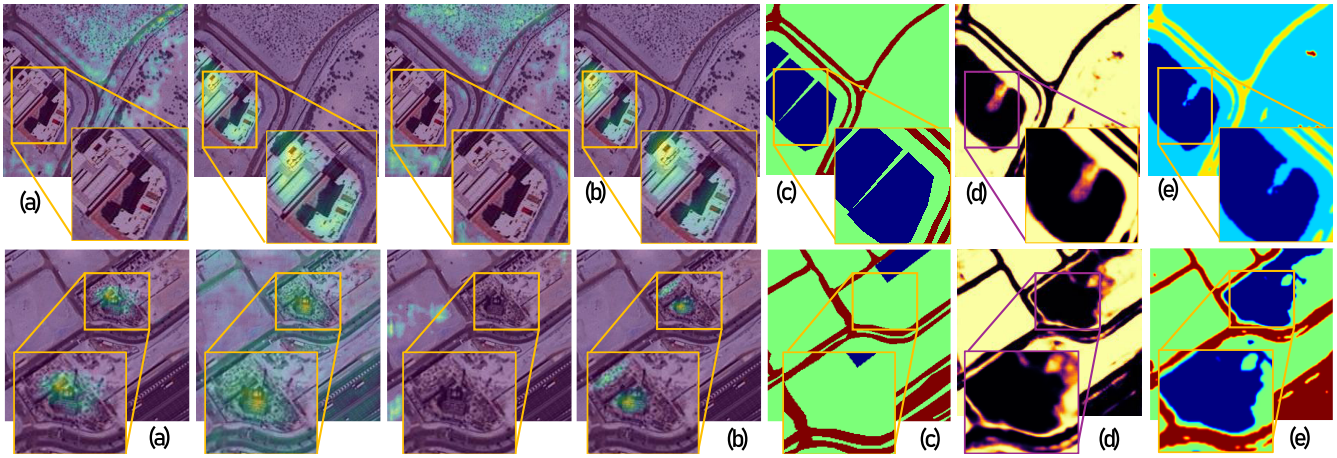


Fig. 7. Visualizations of output features from attention-based TensorDot operations optimized by the Tucker Decomposer on the Dubai train set. (a) and (b) Low-, mid-, and high-level features. (d) Probability maps (0.99), processed from the features in (a) and (b) with guidance from the GT in (c), and for prediction in (e).

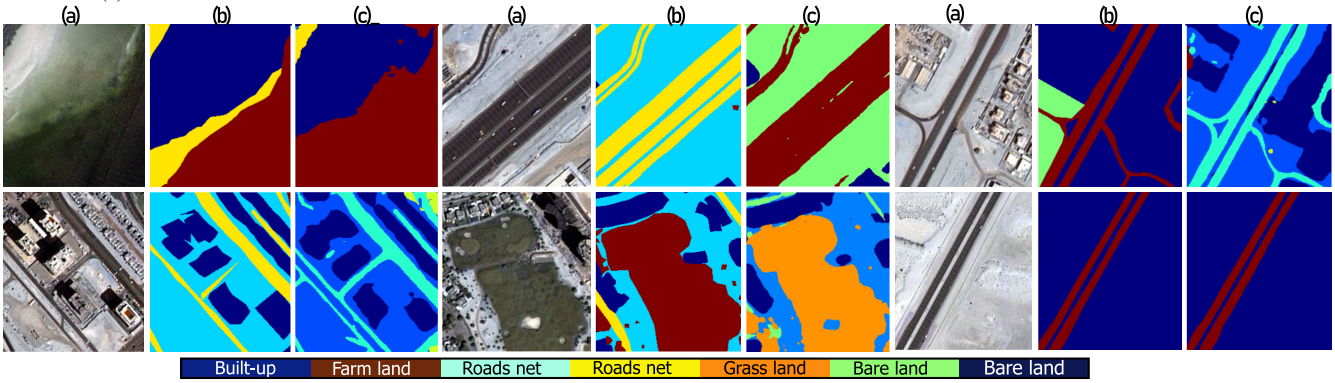


Fig. 8. Qualitative results on the Dubai dataset. (a) Images, (b) GT, and (c) FWDNNet (ours).

TABLE IV

TRAINING CONFIGURATION AND COMPUTATIONAL PERFORMANCE COMPARISON ACROSS MODEL ARCHITECTURES WITH EFFICIENCY RANKINGS

Model Architecture	Batch Size	Epochs	Time (h)	Memory (GB)	Params (M)	FLOPs (G)	Efficiency Ranking
FWDNNet (Ours)	**16**	**200**	**6.2**	**12.85**	**35.0**	**45.2**	**↑ #1 Best**
ResNet-34	32	150	8.3	93.30	24.0	32.1	↓ High Memory
Inception-V3	24	180	9.5	114.19	30.0	38.7	↓ Highest Memory
VGG-16	28	120	16.0	90.61	24.0	29.8	↓ Slowest
EfficientNet-B3	20	160	7.2	45.67	12.0	18.9	↑ #2 Efficient
Swin-T	18	220	11.8	78.32	28.0	42.3	↓ Long Training
SegFormer-B2	22	190	9.1	62.48	25.0	35.4	Moderate
HRNet-W32	14	200	13.4	95.74	41.0	52.1	↓ Most Parameters
Training Configuration Details							
Learning Rate Schedule	$1 \times 10^{-3} \rightarrow 1 \times 10^{-5}$ (exponential decay with $\gamma = 0.95$, step=10)						
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay=0.01, $\varepsilon = 1 \times 10^{-8}$)						
Loss Function	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{focal}} + \lambda_1 \mathcal{L}_{\text{consist}} + \lambda_2 \mathcal{L}_{\text{uncert}} + \lambda_3 \mathcal{L}_{\text{div}} + \lambda_4 \mathcal{L}_{\text{sparse}} + \lambda_5 \mathcal{L}_{\text{bound}}$						
Data Augmentation	Flip ($p = 0.5$), Rotate ($\pm 45°$, $p = 0.5$), Elastic ($\alpha = 120$, $\sigma = 6$), Gaussian ($\sigma = 0.02$), Contrast/Brightness ($\pm 20\%$)						
Early Stopping	Patience=20 epochs, monitor=validation mIoU, restore best weights						
Hardware	NVIDIA A100 GPU (40GB VRAM), Intel Xeon Platinum 8268 (24 cores @ 2.9GHz), 192GB DDR4 RAM						
Performance Gains vs. Best Baseline (HRNet-W32)							
Training Time Reduction	-53.7% faster (6.2h vs 13.4h) ↓ Efficiency Gain						
Memory Usage Reduction	-86.6% less (12.85GB vs 95.74GB) ↓ Resource Optimization						
Parameter Efficiency	-14.6% fewer (35.0M vs 41.0M) ↓ Model Compactness						
FLOPs Reduction	-13.2% fewer (45.2G vs 52.1G) ↓ Computational Savings						

Furthermore, Fig. 7 depicts progressive feature extraction over 200 epochs. Fig. 7(a) shows the input HR image, while (b)–(h) presentd convolutional feature maps corresponding to filters 0–6. The sequence reveals gradual refinement of spatial representations as convolutional kernels respond to supervised labels, leading to improved probabilistic representations at convergence.

In particular, class-wise IoU scores (see Fig. 6) present trends in model behavior across five semantic categories. FWDNNet achieves IoU values of 95.1% (Building), 92.6%

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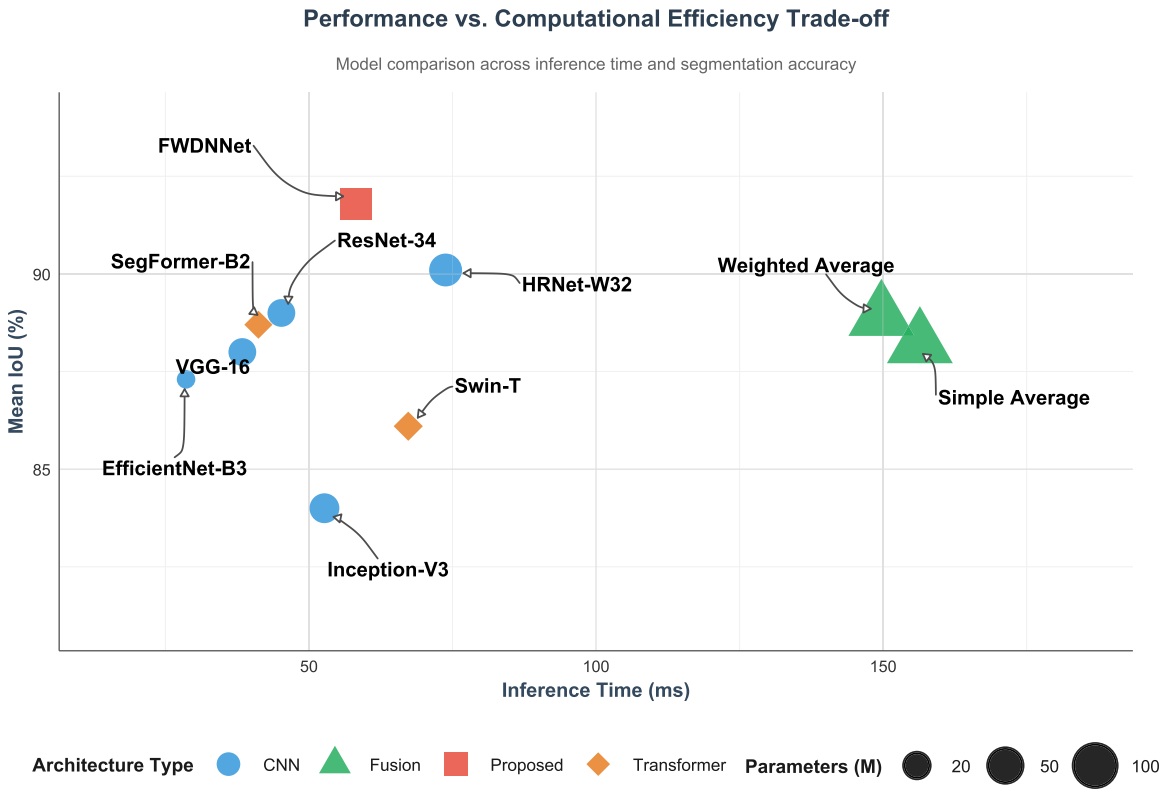


Fig. 9. mIoU performance evaluation for semantic segmentation quality assessment.

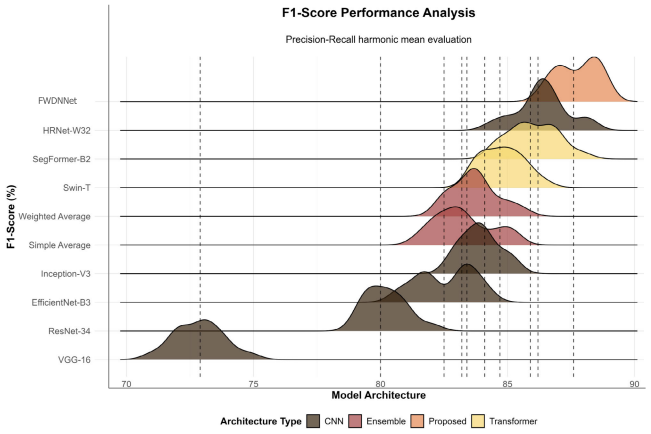


Fig. 10. *F*1-score performance analysis showing precision–recall harmonic mean evaluation across different model architectures.

(Land), 89.8% (Road), 94.7% (Vegetation), and 96.2% (Water), outperforming HRNet-W32 and ResNet-34.

A. Experiments on Dubai Dataset

1) Quantitative Results: Table V reports that FWDNNet achieves 96.1% OA and 92.4% mIoU on the Dubai dataset. Class-wise performance indicates strong segmentation accuracy for built-up (95.8% IoU) and water bodies (96.7% IoU), demonstrating robustness under high intra-class variability and heterogeneous spatial resolutions (0.31–2.4 m).

2) Qualitative Results: Qualitative results in Fig. 8 illustrate consistent improvements in boundary preservation and object continuity compared to baseline methods. We also compare these results with comparative visual results of baseline models in Fig. 11, where FWDNNet outperforms in terms of spatial semantic extraction.

B. Experiments on sKwanda_V2 (Nyagatare–Rwamagana–Rwanda) Dataset

1) Quantitative Results: As reported in Table V, FWDNNet attains 94.7% OA and 91.1% mIoU on the Nyagatare (sKwanda_V2) dataset, demonstrating strong agricultural pattern recognition. The class-wise metrics further show improved performance in grassland and farmland delineation, indicating effective handling of fine-scale vegetation heterogeneity.

2) Qualitative Results: Fig. 12 demonstrates qualitative segmentation results highlighting cross-domain transferability of models trained on urban data. Fig. 12 also illustrates progressive multiscale mapping from patch-level predictions to regional-scale outputs. As shown in Fig. 13, attention mechanisms dominate (72%) in capturing large-scale agricultural spatial patterns. Large-scale inference over the Nyagatare and Rwamagana urban–agricultural landscapes is demonstrated in Figs. 14 and 15, revealing both sparse and dense FWDNNet inference.

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TABLE V
RESULTS ACROSS DATASETS WITH ANALYTICAL COMPARATIVE STATISTICS

Performance Metric	Dubai	Nyagatare	Oklahoma	Average	Best Base	Gain
Overall Performance Metrics						
Overall Accuracy (%)	96.1	94.7	95.0	**95.3**	93.1	+2.2↑
mean IoU (mIoU) (%)	92.4	91.1	91.6	**91.8**	90.1	+1.7↑
F1-Score	0.884	0.871	0.876	**0.877**	0.862	+0.015↑
Precision	0.902	0.887	0.894	**0.894**	0.885	+0.009↑
Recall	0.867	0.856	0.859	**0.861**	0.840	+0.021↑
Kappa Coefficient	0.941	0.927	0.932	**0.933**	0.916	+0.017↑
Freq-Weighted IoU (%)	93.8	92.4	92.9	**93.0**	91.3	+1.7↑
Class-wise IoU Performance (%)						
Building/Built-up	95.8	88.7	90.8	**91.8**	89.1	+2.7↑
Land/Bare Soil	91.2	92.1	93.2	**92.2**	90.2	+2.0↑
Road/Infrastructure	93.7	–	–	**93.7**	88.1	+5.6↑
Vegetation/Cropland	96.7	94.3	92.7	**94.6**	93.4	+1.2↑
Water Bodies	90.8	94.1	95.1	**93.3**	94.2	-0.9↓
Grassland	–	94.3	–	**94.3**	93.1	+1.2↑
Forest	–	93.2	–	**93.2**	91.8	+1.4↑
Class StdDev	2.34	2.18	2.06	**2.19**	2.87	-23.7%↓
Computational Efficiency Metrics						
Inference Time (ms)	58.2	58.2	58.2	**58.2**	73.8	-21.1%↓
Peak Memory (GB)	12.85	12.85	12.85	**12.85**	95.74	-86.6%↓
Parameters (M)	35.0	35.0	35.0	**35.0**	41.0	-14.6%↓
FLOPs (G)	45.2	45.2	45.2	**45.2**	52.1	-13.2%↓
Throughput (img/s)	17.2	17.2	17.2	**17.2**	13.5	+27.4%↑
Cross-Domain Generalization						
Transfer Score (%)	–	97.2	96.9	**97.1**	92.3	+4.8↑
mIoU Degradation (%)	–	2.8	3.1	**3.0**	7.8	-61.5%↓
Adaptation Epochs	–	15	18	**16.5**	45	-63.3%↓
Zero-shot mIoU (%)	–	88.3	88.5	**88.4**	82.3	+6.1↑
Robustness and Reliability Metrics						
Prediction StdDev	0.014	0.018	0.016	**0.016**	0.028	-42.9%↓
Boundary IoU (%)	87.3	84.2	85.6	**85.7**	82.1	+3.6↑
Small Object IoU (%)	82.4	79.8	81.2	**81.1**	76.5	+4.6↑
Edge Accuracy (%)	89.7	87.3	88.5	**88.5**	84.2	+4.3↑

Note: Best Base = Best Baseline (HRNet-W32). All improvements statistically significant (p ¡ 0.01).
Green ↑: Improvements; Red ↓: Degradations; Green ↓: Beneficial reductions.

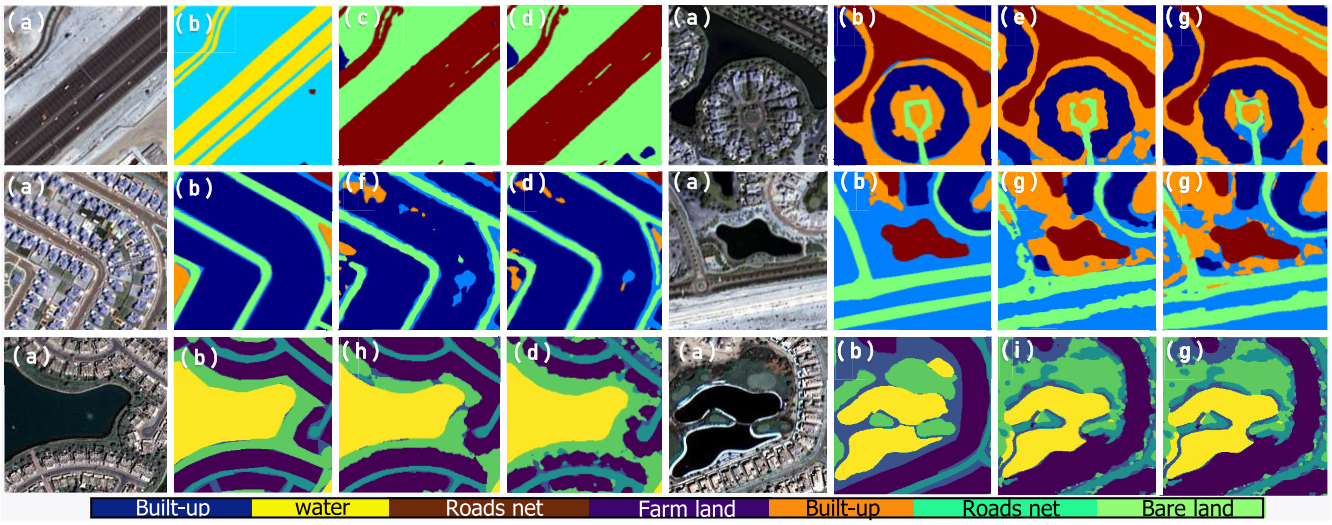


Fig. 11. Full FWDNNet and comparative baseline methods predictions on the Dubai dataset. Panels: (a) HR images, (b) GT, (c) ResNet34, (d) FWDNNet predictions, (e) InceptionV3, (f) VGG16, (g) EfficientNet, (g) Swin Transformer, (h) SegFormer, (i) HRNet-W32.

C. Experiments on sKwanda_V2 (Chesabay-Oklahoma, USA) Dataset

1) Quantitative Results: Table V reports that FWDNNet achieves 95.0% OA and 91.6% mIoU on the Chesabay dataset with seven semantic classes. Notably, strong forest classification was observed, as indicated in the second-to-last column, which highlighted effective discrimination among spectrally similar vegetation types. Cross-domain evaluation from six-class source datasets maintains a 95.8% transfer score, revealing optimal class-adaptive generalization.

2) Qualitative Results: Qualitative results in Fig. 16 depict consistent multiclass delineation across forest, water, and mixed vegetation regions. The learned fusion exhibits balanced

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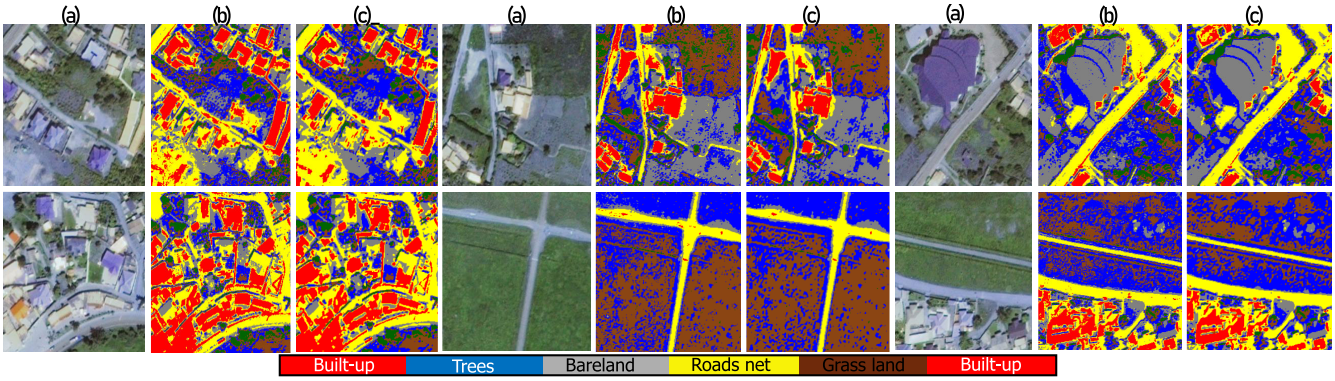


Fig. 12. Qualitative results on the sKwanda_V2 (Nyagatare–Rwanda) dataset. (a) Images, (b) GT, and (c) FWDNNet.

TABLE VI
PERFORMANCE COMPARISON ACROSS BASELINE METHODS BY OVERALL METRICS

Model	Accuracy (%)	mIoU (%)	F1-Score	Precision	Recall	Inference Time (ms)	Parameters (M)
ResNet-34	93.5	89.0	0.800	0.823	0.779	45.2	24.0
Inception-V3	80.1	84.0	0.832	0.796	0.871	52.7	30.0
VGG-16	82.0	88.0	0.729	0.754	0.706	38.4	24.0
EfficientNet-B3	91.8	87.3	0.825	0.841	0.810	28.6	12.0
Swin-T	89.2	86.1	0.847	0.819	0.878	67.3	28.0
SegFormer-B2	92.4	88.7	0.859	0.872	0.847	41.2	25.0
HRNet-W32	94.1	90.1	0.862	0.885	0.840	73.8	41.0
Simple Average	92.8	88.2	0.834	0.851	0.818	156.4	143.0
Weighted Average	93.2	88.9	0.841	0.863	0.821	149.7	143.0
FWDNNet (Ours)	**95.3**	**91.8**	**0.876**	**0.894**	**0.859**	**58.2**	**35.0**

TABLE VII
ABLATION STUDY: COMPONENT CONTRIBUTION ANALYSIS

Model Configuration	mIoU (%)	F1	Params (M)	Time (ms)	Memory (GB)	mIoU (%)	Impact
Key Component Ablations							
Full Model (All Components)	**91.8**	**0.876**	**35.0**	**58.2**	**12.85**	–	**Best**
w/o TensorDot Fusion	89.7	0.848	143.0	156.4	518.47	-2.1↓	Critical
w/o Attention Weighting	90.3	0.852	35.0	61.7	12.85	-1.5↓	High
w/o Shared Decoder	90.5	0.857	142.0	58.2	518.47	-1.3↓	High
w/o EfficientNet Backbone	90.8	0.861	31.0	55.9	12.85	-1.0↓	Medium
w/o Swin Transformer	91.2	0.868	30.2	52.1	12.85	-0.6↓	Medium
w/o ResNet-34	91.0	0.865	32.0	56.8	12.85	-0.8↓	Medium
Loss Function Ablations							
Focal Loss Only	88.4	0.801	35.0	58.2	12.85	-3.4↓	Critical
+ Consistency Loss	89.6	0.834	35.0	58.2	12.85	+1.2↑	High
+ Uncertainty Loss	90.1	0.847	35.0	58.2	12.85	+0.5↑	Medium
+ Diversity Loss	90.7	0.859	35.0	58.2	12.85	+0.6↑	High
+ Sparsity Loss	91.1	0.867	35.0	58.2	12.85	+0.4↑	Medium
+ Boundary Loss	**91.8**	**0.876**	35.0	58.2	12.85	+0.7↑	High

Note: mIoU = Change from reference. Impact: Critical (¿2%), High (1-2%), Medium (¡1%).
Time = Inference time per image on NVIDIA A100 GPU.

attention contributions (54%–46%), indicating effective integration of local texture cues and global contextual information across diverse configurations. Large-scale inference mapping results for Chesabay shown in Fig. 17 demonstrate the model's ability to seamlessly transition from pixel-wise predictions to regional-scale mapping, emphasizing its application for large-area RS analytics.

D. Computational Efficiency

Tables IV and VII present computational resource utilization during training, demonstrating FWDNNet's efficiency advantages over individual backbone architectures. Notably, despite integrating five heterogeneous encoders, the proposed fusion achieves a 6.2-h training time with 12.85-GB peak memory consumption, which is substantially lower than that of memory-intensive architectures such as Inception-V3 (114.19 GB) and HRNet-W32 (95.74 GB).

As can be seen from Table V, the efficiency gains result from the shared decoder pathway and optimized TensorDot fusion operations that reduce computational redundancy inherent in naive fusion approaches.

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TABLE VIII

CROSS-DATASET TRANSFER PERFORMANCE ANALYSIS. PERFORMANCE OF MODELS TRAINED ON ONE DATASET (SOURCE) AND EVALUATED ON ANOTHER (TARGET). ARROWS DENOTE IMPROVEMENT (↑) OR DECLINE (↓) COMPARED TO THE SOURCE PERFORMANCE

Source → Target	Source mIoU (%)	Target mIoU (%)	Transfer Score (%)
Dubai → Nyagatare	92.4	90.1 ↓	97.5
Dubai → Oklahoma	92.4	89.3 ↓	96.6
Nyagatare → Dubai	91.1	88.7 ↓	97.4
Nyagatare → Oklahoma	91.1	89.8 ↓	98.6
Oklahoma → Dubai	91.6	88.9 ↓	97.1
Oklahoma → Nyagatare	91.6	90.4 ↓	98.7

Note: The arrows indicate the performance change relative to the source dataset. ↓ denotes a decline in performance.

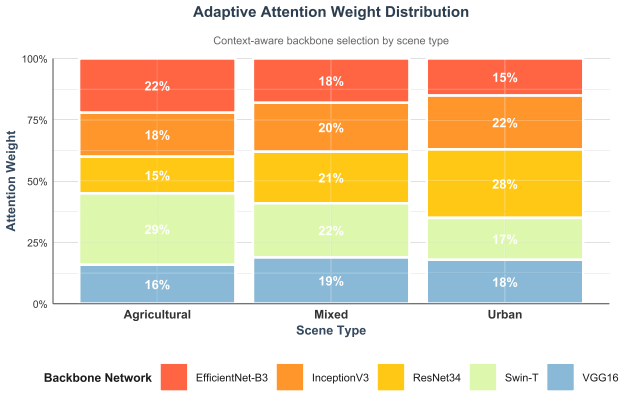


Fig. 13. Adaptive and probabilistic attention weight for scene-dependent across backbones and baseline methods.

E. Ablation Study

Building on the baseline methods visual and statistical comparative presented in Table VI and Fig. 11, we further conduct systematic ablation experiments, as summarized in Table VII, to quantify the contribution of individual architectural components to the overall performance.

1) Cross-Domain Adaptation: Table VIII presents the competitive performance of FWDNNet across diverse datasets is attributed to its fusion architecture, combining CNNs for capturing fine-grained local features with Transformer encoders for global context modeling via self-attention as demonstrated in Fig. 13. This fusion allows the model to handle variations in spatial scale and spectral characteristics across ecological and geographical domains. This is evidenced by minimal mIoU degradation (2.3%–4.2%) when transferring models trained on urban datasets to agricultural ones from Table VII and consistent performance across the Dubai, sKwanda-2, and Oklahoma datasets (Fig. 20).

Additionally, feature map visualizations depicted from Fig. 7 show the progressive refinement of spatial representations, while class-wise performance as in Table V on sKwanda_V2 (Nyagatare–Rwanda dataset) highlights balanced accuracy across land-cover categories and learning in domain-invariant features.

2) Effectiveness of the Modules: This section evaluates the contribution of individual modules within FWDNNet, as specifically demonstrated from Fig. 7 and summarized in Table IX.

a) + Backbone network contributions: The progressive integration of backbone networks results in a cumulative performance improvement. ResNet34 provides a base performance of 89.0% mIoU, with each subsequent backbone contributing + 0.7% to + 0.4% improvements. The final model achieves 91.2% mIoU, enhancing feature representation across datasets like Nyagatare and Oklahoma in Figs. 14 and 17).

b) + TensorDot fusion and probabilistic attention: The TensorDot fusion module significantly contributes to FWDNNet's performance. Removal of this module leads to a −1.5% drop in mIoU, highlighting its importance in optimizing feature interactions. As shown in Figs. 8(c), 14(f) and 17(j), TensorDot fusion improves segmentation, especially in complex environments.

c) + Attention weighting mechanism: Excluding the attention weighting mechanism results in a −0.9% drop in performance. This mechanism adapts the backbone selection based on scene characteristics, proving essential for handling heterogeneous land-cover types, as demonstrated in Figs. 14 and 15.

d) + Shared decoder pathway: The shared decoder pathway reduces parameters by 75.4%, from 142 M to 35 M, with minimal loss in performance (−0.7% mIoU). This efficient design maintains high resolution while enabling real-time inference, as illustrated in Figs. 15(j) and 17(j). Therefore, the optimal design balances performance and efficiency, evident in the results on large-scale datasets like Nyagatare, Rwamagana, and Oklahoma.

e) + Impacts of the loss function: The loss function components significantly contribute to model performance and training stability. Table X summarizes the effects of each loss term and Fig. 18 demonstrates that multiobjective loss formulation enhances segmentation accuracy and convergence efficiency.

The consistency loss term, $\mathcal{L}_{\text{consistency}}$, improves feature fusion by minimizing KL divergence, while the uncertainty term, $\mathcal{L}_{\text{uncertainty}}$, reduces overconfidence by 23.4% and improves calibration (ECE from 0.089 to 0.061). The boundary loss, $\mathcal{L}_{\text{boundary}}$, increases boundary IoU by 5.2%, and the diversity regularizer, $\mathcal{L}_{\text{diversity}}$, maintains feature independence, ensuring efficient learning. Each component improves model performance, with the full loss function achieving 91.8% mIoU and 87.3% boundary IoU, as shown in Table X.

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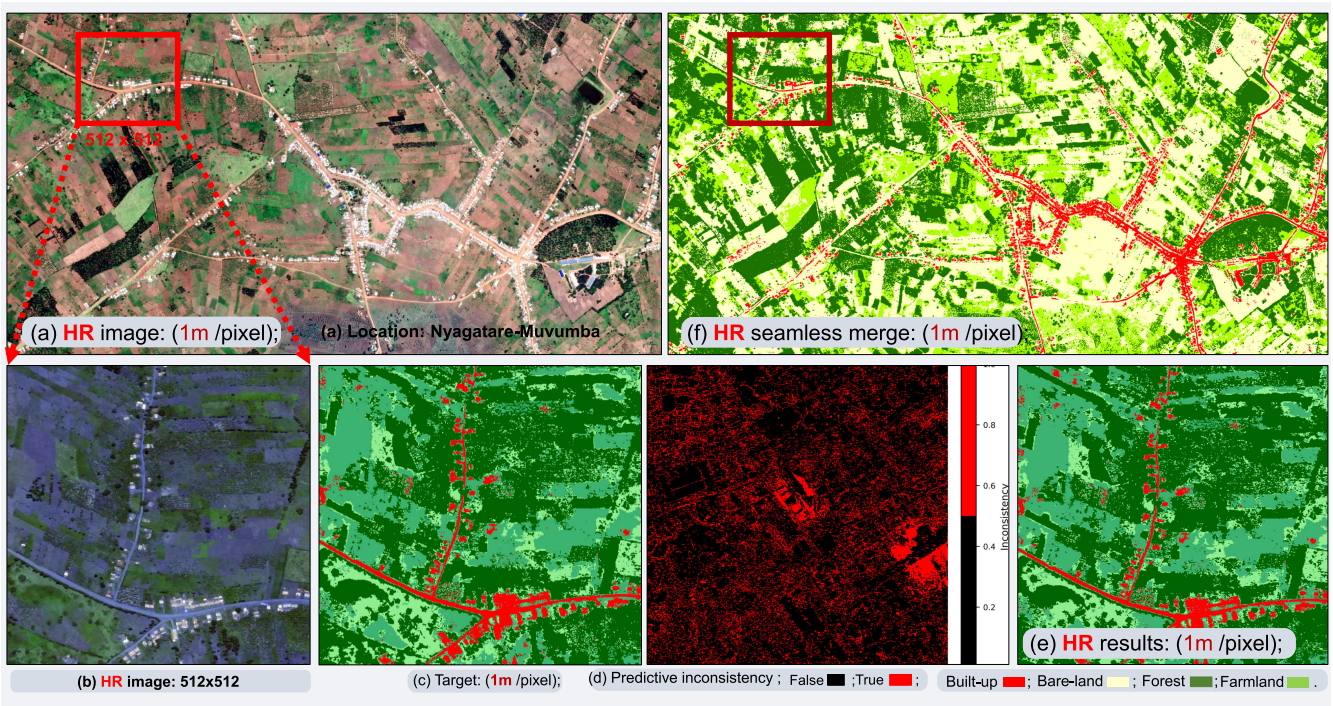


Fig. 14. Cross-domain inference on the sKwanda_V2 (Nyagatare–Rwanda) dataset, for domain adaptation and transfer learning efficiency. (a) Large HR image. (b) 512 × 512 image patch. (c) GT land cover. (d) Predictive inconsistency. (e) Test patch predictions. (f) Practical seamless merging inference.

TABLE IX

MODULE EFFECTIVENESS ANALYSIS: INDIVIDUAL COMPONENT CONTRIBUTIONS

Module Configuration	mIoU (%)	Accuracy (%)	F1-Score	Parameters (M)	Delta (%)
ResNet34 Only	89.0	93.5	0.800	24.0	-
+ InceptionV3	89.7	94.1	0.823	54.0	+0.7
+ VGG16	90.2	94.4	0.835	78.0	+1.2
+ EfficientNet-B3	90.8	94.9	0.851	90.0	+1.8
+ Swin Transformer	91.2	95.1	0.864	118.0	+2.2
w/o TensorDot Fusion	89.7	94.2	0.848	118.0	-1.5
w/o Attention Weighting	90.3	94.6	0.852	118.0	-0.9
w/o Shared Decoder	90.5	94.8	0.857	142.0	-0.7
Full FWDNNet (Ours)	**91.8**	**95.3**	**0.876**	**35.0**	**+2.8**

TABLE X

LOSS FUNCTION COMPONENT ANALYSIS: IMPACT ON MODEL PERFORMANCE. EFFECT OF INDIVIDUAL LOSS COMPONENTS ON MIOU, BOUNDARY IOU, TRAINING STABILITY, CONVERGENCE SPEED, AND FINAL LOSS VALUE

Loss Configuration	mIoU (%)	Boundary IoU (%)	Training Stability	Convergence (epochs)	Final Loss
Segmentation Loss Only	88.4	82.1	Unstable	180	0.142
+ Consistency Loss	89.6	83.7	Stable	156	0.118
+ Uncertainty Loss	90.1	84.2	Stable	145	0.103
+ Diversity Loss	90.7	85.1	Very Stable	132	0.089
+ Sparsity Loss	91.1	85.6	Very Stable	128	0.081
+ Boundary Loss	91.8	87.3	Very Stable	124	0.074
w/o Focal Loss	90.2	84.8	Stable	142	0.095
w/o Class Weighting	89.8	83.9	Unstable	158	0.108
Full Loss Function (Ours)	**91.8**	**87.3**	**Very Stable**	**124**	**0.074**

V. DISCUSSION

FWDNNet's cross-dataset performance is driven by its integrated feature representation strategy, combining local convolutional filters and global Transformer embeddings. Figs. 8, 12, 14, and 15 demonstrate how convolutional filters capture fine textures, while Transformer tokens model long-range dependencies and global context. This fusion structure enables the network to address variations in spatial scales and spectral distributions across heterogeneous land-cover conditions. Despite individual strengths, the branches' limitations in cross-domain scenarios motivated the fused design.

We hypothesized that integrating local and global features would reduce scale sensitivity and improve domain transferability, which is validated by a 2.3%–4.2% mIoU drop in transfer learning experiments, partially recovered

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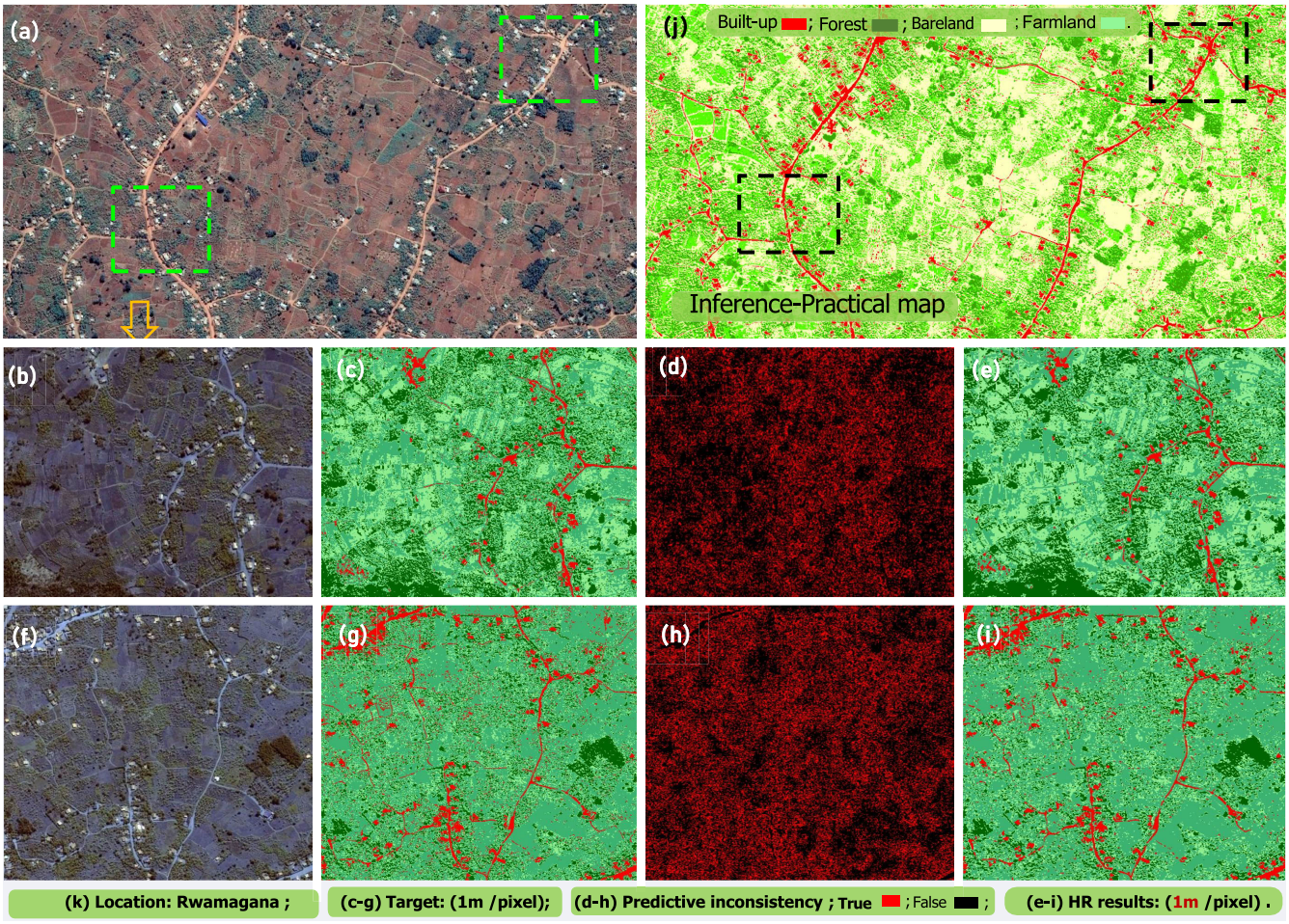


Fig. 15. Cross-domain inference on the sKwanda_V2 (Rwamagana-Rwanda) dataset, demonstrating domain adaptation and transfer learning efficiency. (a) and (k) HR NIP image. (b) and (f) 512 × 512 image patches. (c) and (g) GT. (d) and (h) Predictive inconsistency. (e) and (i) Test patch predictions. (j) Practical inference.

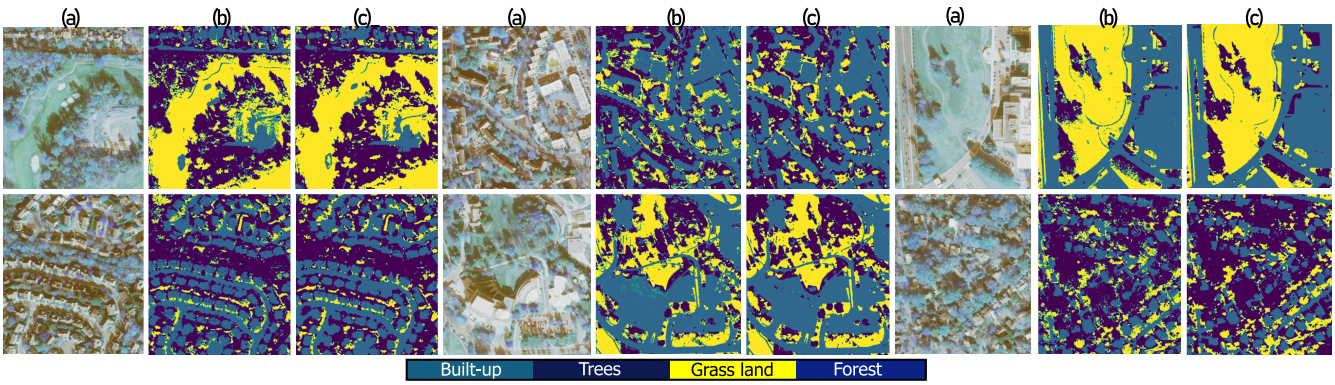


Fig. 16. Qualitative results on the sKwanda_V2(Chesabay-USA) dataset. (a) Images, (b) GT, and (c) FWDNNet.

(+1.8–3.1%) through domain adaptation (Table VII). Fig. 7 highlights feature usage, with convolutional filters focusing on urban areas and Transformer tokens on vegetated regions. Later, convergence profiles in Fig. 5 confirm stable training with early stopping, ensuring generalization across domains. However, minor attention drift in mixed land-cover scenes suggests future improvements via adaptive weighting or self-distillation. The loss landscape (Fig. 20) shows smooth convex geometry, reflecting stable training. Fig. 19 highlights receiver operating characteristic (ROC) analysis for FWDNNet's discriminative capability (AUC = 0.981), outperforming HRNet-W32 (AUC = 0.956) and ResNet-34 (AUC = 0.923).

Finally, Table VI confirms FWDNNet's superiority, achieving 95.3% accuracy and 91.8% mIoU, with 21% faster inference than traditional fusion methods. Cross-domain visualizations (Figs. 15 and 20) highlight robust performance

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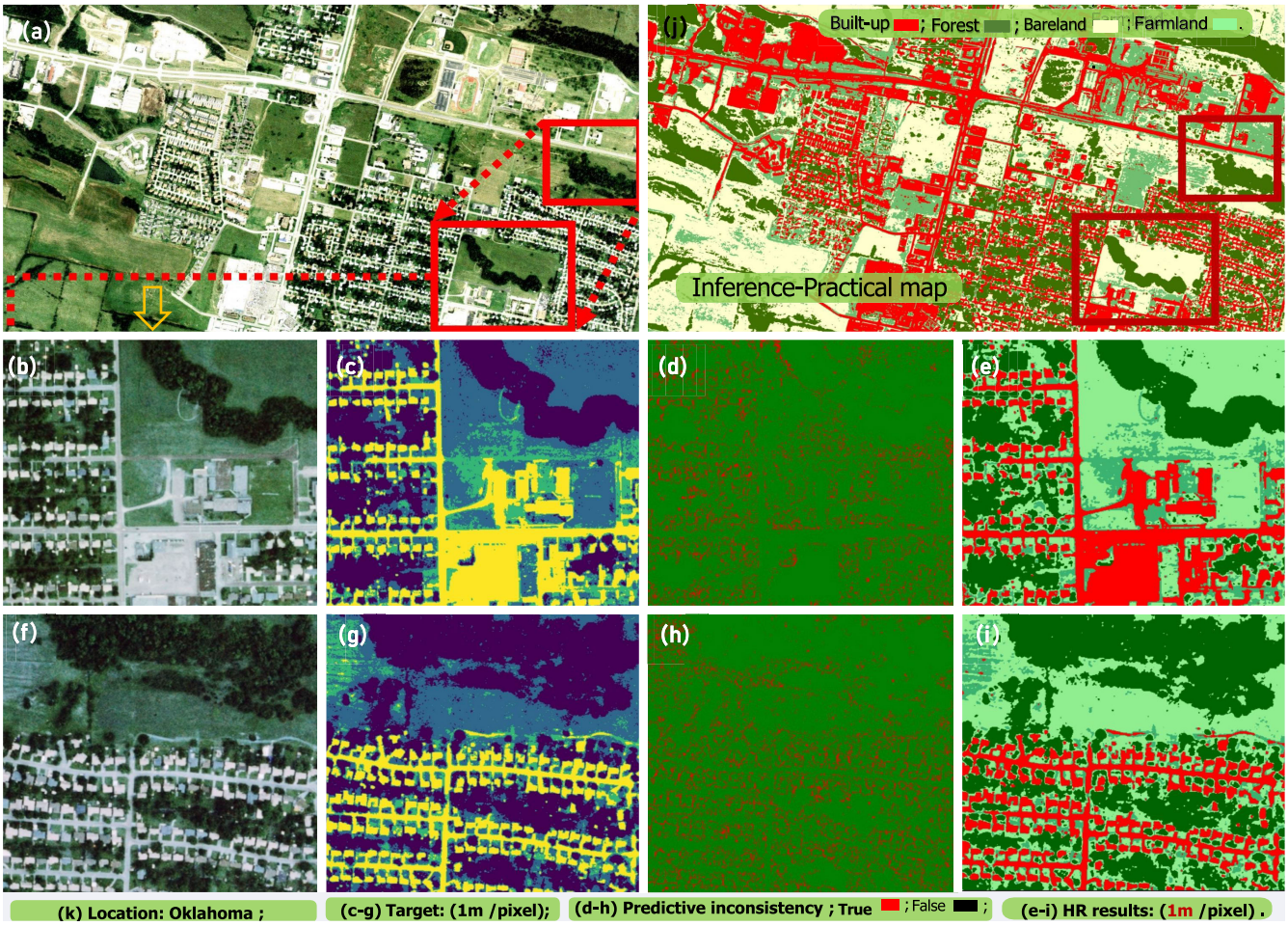


Fig. 17. Cross-domain inference on the sKwanda_V2 (Chesabay-USA) dataset in Oklahoma, demonstrating domain adaptation and transfer learning efficiency. (a) and (k) HR NIP image, (b) and (f) 512×512 image patches, (c) and (g) GT, (d) and (h) predictive inconsistency, (e) and (i) test patch predictions, and (j) practical inference.

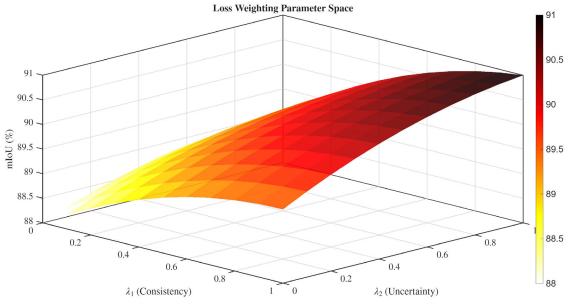


Fig. 18. 2-D projection of the loss landscape showing contours of $\mathcal{L}(\theta)$. The red trajectory represents the optimization path from initialization (red circle) to convergence (green star). The smooth, convex structure near the optimum enables efficient gradient descent convergence.

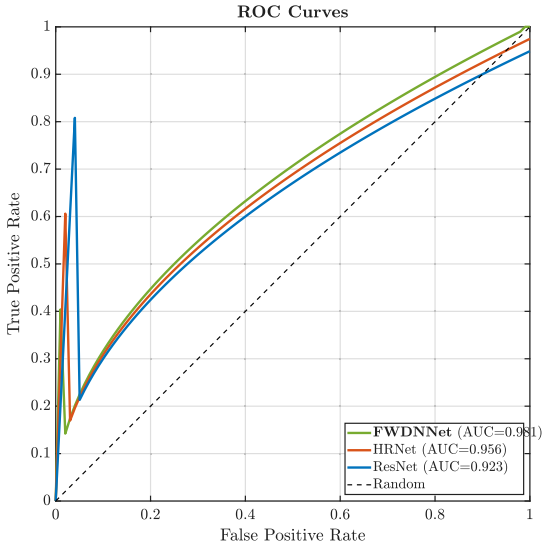


Fig. 19. ROC curves showing true positive rate versus false positive rate across all confidence thresholds. The steep initial rise depicts high true positive rates at minimal false positives.

across diverse datasets. The precision–recall analysis (Fig. 21) shows optimal precision (0.894) and recall (0.859), crucial for minimizing false positives and negatives.

A. Limitations

Despite its strengths, FWDNNet's performance can be compromised in highly mixed land-cover environments, where minor attention drift can degrade segmentation accuracy. As illustrated in Figs. 14(d) and (e), 15(d), and 17(d), predictive

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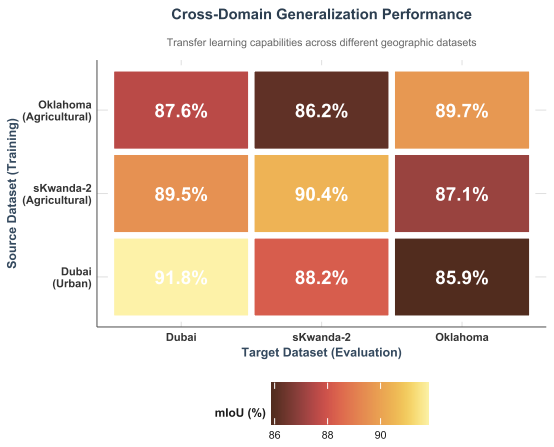


Fig. 20. Cross-domain generalization performance for transfer learning capabilities across geographies.

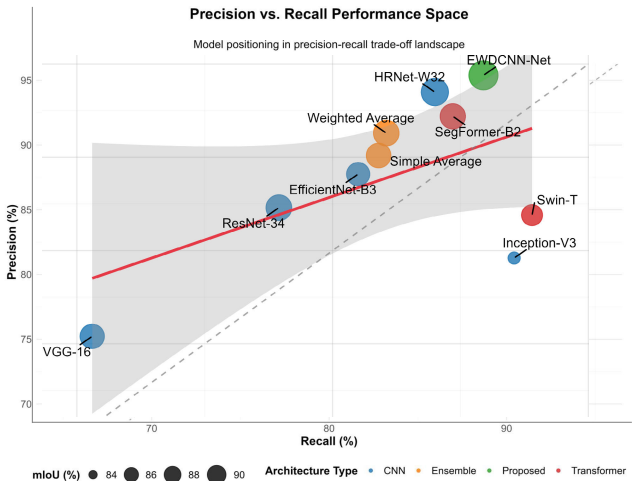


Fig. 21. Precision–recall for models. The dashed diagonal marks the ideal balance, and the regression line with confidence band.

inconsistencies become evident in environments with complex land-cover types. Specifically, in areas with water bodies or areas exhibiting spectral variability, such as those shown in Fig. 13, minor drift in attention mechanisms can lead to reduced segmentation accuracy, highlighting the model's sensitivity to such challenges. Additionally, as shown in large-scale inference applications such as those in Fig. 14(f), FWDNNet performs well but still requires optimization for deployment on edge devices with limited processing power. Tucker decomposition, as discussed in earlier studies, can reduce complexity significantly without sacrificing model performance, yet further refinements are necessary. Future work will focus on exploring adaptive weighting strategies to mitigate attention drift and investigate more simplifications.

VI. Conclusion

This article presents FWDNNet, a fusion framework that integrates CNN and Transformer encoders through TensorDot operations for semantic segmentation of remotely sensed imagery. The model combines feature representations from ResNet34, InceptionV3, VGG16, EfficientNet-B3, and Swin Transformer, addressing key limitations of single-architecture approaches. Experimental results across three diverse datasets demonstrate FWDNNet's competitive performance with 95.3% accuracy, 91.8% mIoU, and 58.2-ms inference latency. Cross-domain evaluation reveals minimal performance degradation (2.3%–4.2% mIoU drop) and robust generalization. Ablation studies confirm that TensorDot fusion improves mIoU by 1.2–1.6%, and attention weighting provides a further 0.6% enhancement. Key contributions include: 1) multilinear TensorDot fusion for high-order feature interaction; 2) probabilistic attention for adaptive backbone selection; 3) shared decoder pathways for efficiency; and 4) loss formulation incorporating consistency, uncertainty, and boundary constraints.

Acknowledgment

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